

Topology-Optimal Multiple Gossip Steps for Decentralized Federated Learning via Gossip Tensor

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Abstract—Decentralized federated learning (DFL) has gained significant attention as a framework for analyzing large-scale data distributed across multiple sites, where communication between sites is constrained by a decentralized graph structure. Due to privacy concerns and high communication costs, reducing the number of communication rounds in DFL has become an important area of research. This article investigates the effects of the decentralized graph's topology on the convergence rate of DFL algorithms and introduces a novel tensor-based multiple-gossip-steps (T-MGS) method to optimize communication efficiency from the topology aspect. The core idea of this method is to use gossip tensors to guide the information flow between sites and enable dynamic adjustments to the transmitted content at each communication step without increasing its volume. The proposed method minimizes the second-largest absolute eigenvalue of the equivalent gossip matrix, a key factor influencing convergence speed. Experimental results on both simulated and real datasets demonstrate that the proposed T-MGS outperforms existing strategies in terms of communication efficiency, reducing the number of communication rounds without compromising model accuracy.

Index Terms—Communication efficiency, communication round, decentralized graph, federated learning, second-largest absolute eigenvalue.

I. INTRODUCTION

FEDERATED learning (FL) has gained significant attention as a framework for analyzing big data, where data are stored across multiple sites and learning models are created by carefully designing the information exchanged between these sites under privacy and communication constraints [1], [2], [3]. Traditional FL approaches rely on a central site that communicates with each participating site to aggregate model updates. However, this centralized architecture poses several issues, including communication bottlenecks, single points of failure, and privacy concerns at the central site [4]. In response, decentralized federated learning (DFL) has emerged as an

TABLE I
RELATIONSHIP BETWEEN μ AND THE REQUIRED COMMUNICATION ROUNDS TO ACHIEVE ε -ACCURACY FOR DFL ALGORITHMS WITH SPECIFIC LOSS FUNCTION SELECTIONS. THE RATES PERTAIN TO SYMMETRIC DOUBLY STOCHASTIC GOSSIP MATRICES WITHOUT EMPLOYING THE MGS TECHNIQUE. TERMS UNRELATED TO μ AND ε ARE OMITTED

Algorithm	# Communication	Loss Function
DGD [9]	$O\left(\frac{\log(1/\varepsilon)}{(1-\mu)^2}\right)$	Strongly convex
Acc-DNGD [10]	$O\left(\frac{\log(1/\varepsilon)}{(1-\mu)^{3/2}}\right)$	Strongly convex
EXTRA [11]	$O\left(\frac{\log(1/(\varepsilon-\varepsilon\mu))}{1-\mu}\right)$	Strongly convex
NIDS [12]	$O\left(\frac{\log(1/\varepsilon)}{1-\mu}\right)$	Lower semicontinuous convex
DFedAvgM [7]	$O(\varepsilon^{-2} + \frac{\varepsilon^{-2/3}}{(1-\mu)^{4/3}})$	Arbitrary
Network-DANE [13]	$O\left(\frac{1/\mu+1}{(1-\mu)^2} \log(1/\varepsilon)\right)$	Quadratic
	$O\left(\frac{\log(1/\varepsilon)}{(1-\mu)^2}\right)$	Arbitrary

alternative, where participating sites communicate with one another following a decentralized graph topology, eliminating the need for a central coordinator and making the system more robust and scalable for real-world applications [4].

Without a central site, DFL encounters a significant challenge in communication efficiency. In decentralized graphs, each site must communicate with its neighbors over multiple rounds to propagate updates across distant sites, thereby significantly increasing the number of required communication rounds. In scenarios where each communication round is costly—such as cross-border collaborations between companies requiring government data review before transmission [5]—an excessive number of rounds can lead to high operational costs and considerable delays. Therefore, minimizing communication rounds without sacrificing model accuracy is an important task in DFL.

Many studies have examined the impact of a decentralized graph's topology on the communication efficiency of DFL algorithms [6], [7], [8]. The gossip matrix, a weighted adjacency matrix of the decentralized graph, guides the consensus of information received from other sites after each communication round of DFL algorithms. Notably, the second-largest absolute eigenvalue of the gossip matrix has been widely recognized as a key factor influencing the convergence rate. A higher second-largest absolute eigenvalue (denoted by μ) results in slower convergence, requiring more communication rounds to achieve a given level of accuracy. Table I sum-

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marizes the relationship between μ and the required number of communication rounds for typical DFL algorithms. Given the presence of the factor $1 - \mu$ in the denominator of communication rounds, reducing μ is critical for minimizing communication overhead and optimizing efficiency in DFL systems.

Several methods have been proposed to mitigate the effects of μ on convergence speed. For instance, specific graph structures, such as exponential graphs [14], hypercube graphs [15], and dynamic graphs [16], [17], are designed to minimize μ . However, in many real-world scenarios, the communication graph is inflexible and determined by the collaboration patterns between sites, making it difficult to redesign the graph during the learning process. As a result, optimizing the communication process while maintaining a fixed, deterministic graph structure remains an open challenge. To address this challenge, a promising alternative is using multiple gossip steps (MGSs) [18], [19], where several communication rounds are conducted in each iteration. The key idea behind MGS is that performing MGSs can effectively improve the eigenvalue gap, thereby accelerating the convergence rate. However, while MGS can reduce communication rounds in some cases, it has not demonstrated sufficient improvement, particularly when the graph is sparse and μ is large.

In this article, we focus on deterministic decentralized graphs and propose a novel gossip tensor-based method to enhance MGS and optimize communication efficiency in DFL. Our key innovation lies in the introduction of a gossip tensor, which enables more flexible information transmission between sites. In traditional MGS, the gossip matrix remains static across multiple steps, which does not fully utilize the potential of communication to reduce redundancy and accelerate convergence. In contrast, the proposed method: 1) leverages tensor representations to allow each site to transmit different content to neighbor sites and 2) enables the gossip matrix to change dynamically during each gossip step. These adjustments facilitate more efficient communication by minimizing μ of the equivalent gossip matrix without increasing the volume of content transmitted during MGS, which reaches or approximates a kind of “topology optimality” in DFL. We outline our primary contributions below.

- 1) *Optimized Dynamic Information Transmission Strategy:* We introduce a gossip tensor-based MGS method, an efficient communication strategy in DFL. This new method provides a novel solution for effective data transmission, including optimizing information flow across decentralized networks and dynamically varying content.
- 2) *Efficient Implementation Algorithm:* We develop an efficient algorithm to determine the gossip tensors that minimize μ and accelerate convergence.
- 3) *Theoretical Guarantee:* We provide a theoretical analysis showing that our method outperforms traditional MGS techniques by achieving a lower μ in the equivalent gossip matrix, thereby improving the convergence rate.
- 4) *Numerical Validation:* We validate our method through extensive experiments, showing that it outperforms traditional MGS in reducing communication rounds, particularly for sparse graphs.

The structure of this article is as follows. Section II introduces a general DFL framework encompassing most existing DFL algorithms. Section III discusses the effects of the decentralized graph’s topology on the convergence rate of DFL. In Section IV, we present the proposed framework and the algorithm for determining gossip tensors. Section V provides experimental results on both simulated and real-world datasets. Section VI reviews related works, and Section VII concludes the article.

II. PROBLEM SETTING: A GENERAL DFL FRAMEWORK

Suppose that there are m fixed sites in the problem. We use $\mathcal{D}_i$ to denote the dataset at the i^{th} site. For convenience, we consider the case of equal sample sizes at each site. The communication between sites is guided by an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the vertex set and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the edge set. Each vertex in $\mathcal{V}$ represents a site, and each edge (i, j) indicates that site i can communicate with site j . $\mathcal{G}$ also includes self-loop edges, that is, $(i, i) \in \mathcal{E}$ for any $i \in \mathcal{V}$. Site j is called a neighbor of site i if $(i, j) \in \mathcal{E}$, and we define $\mathcal{N}(i) = \{j : (i, j) \in \mathcal{G}\}$ as the neighbor set of i . Our study focuses on the deterministic decentralized graph, where $\mathcal{G}$ is predefined and remains fixed. We also assume that $\mathcal{G}$ is connected, meaning there is a path between every pair of vertices. $\mathbf{W} = [w_{ij}] \in \mathbb{R}^{m \times m}$ is called a gossip matrix of $\mathcal{G}$ if $w_{ij} \geq 0$ for $(i, j) \in \mathcal{E}$, $w_{ij} = 0$ for $(i, j) \notin \mathcal{E}$, and $\sum_{j=1}^m w_{ij} = 1$ for each $i = 1, 2, \dots, m$. For any gossip matrix $\mathbf{W}$, we use $\lambda_1(\mathbf{W}) \geq \lambda_2(\mathbf{W}) \geq \dots \geq \lambda_m(\mathbf{W})$ to denote the eigenvalues of $\mathbf{W}$. We define $\mu(\mathbf{W})$ as the second-largest absolute eigenvalue of $\mathbf{W}$, that is, $\mu(\mathbf{W}) = \max(|\lambda_2(\mathbf{W})|, |\lambda_m(\mathbf{W})|) \in [0, 1)$.

Use $\theta \in \mathbb{R}^d$ to denote all parameters to be estimated in the problem. The primary goal of DFL is to learn θ by minimizing the following objective function:

$$L(\theta) = \sum_{i=1}^m l(\mathcal{D}_i, \theta) \quad (1)$$

where $l(\cdot)$ is a loss function that evaluates the local performance of θ at each site. If all observations are stored at a single site, (1) reduces to a standard machine-learning problem solvable by gradient-based methods. However, within the DFL framework, $\mathcal{D}_i$ is only accessible at the i^{th} site and cannot be transmitted to other sites. Consequently, the most common strategy is to generate a local estimator θ_i at the i^{th} site and construct an iterative algorithm that sequentially updates, communicates, and aggregates the local estimators during each iteration. After a sufficient number of iterations, a global estimator of θ is obtained at each site by aggregating all information from $\mathcal{D}_1, \dots, \mathcal{D}_m$.

We present a general iterative framework of DFL to outline the above strategy in Algorithm 1.

Algorithm 1 encompasses a series of DFL methods by specifying the form of the updating function f . For instance, if $f(\theta_i^{(t-1)}; \mathcal{D}_i) = \theta_i^{(t-1)} - \eta_t(\partial l(\mathcal{D}_i, \theta))/(\partial \theta)|_{\theta=\theta_i^{(t-1)}}$, with the step size η_t , this framework becomes the DGD algorithm [20].

The communication and consensus step in Algorithm 1 is commonly referred to as one gossip step (1GS) in the literature [18], [19]. During 1GS, each site collects local estimators from its neighbors and creates a consensus estimator through

Algorithm 1 DFL Framework

Input: $\{\mathcal{D}_i\}_{i=1}^m$, $\mathcal{G}$, $\mathbf{W}$, and an initial local estimator $\theta_i^{(0)}$ at each i^{th} site.

for $t = 1, 2, \dots, T$ **do**

I. **Local Updating:** Using an updating function f , calculate

$$\theta_i^{(t-0.5)} \leftarrow f\left(\theta_i^{(t-1)}; \mathcal{D}_i\right), \quad (2)$$

II. **Communication and Consensus (One Gossip Step):** Each site sends $\theta_i^{(t-0.5)}$ to its neighbors $\mathcal{N}(i)$, and

$$\theta_i^{(t)} \leftarrow \sum_{j \in \mathcal{N}(i)} w_{ij} \theta_j^{(t-0.5)}. \quad (3)$$

end for

Output: $\theta_i^{(T)}$ as the final estimate of θ at the i^{th} site.

weighted averaging with w_{ij} . This step facilitates the mixing of information from different sites. Therefore, $\mathbf{W}$ plays a crucial role in DFL. The computational complexity between each pair of connected sites is only $2|\theta|$, where $|\theta|$ represents the number of elements in θ .

III. EFFECTS OF GRAPH TOPOLOGY ON THE CONVERGENCE RATE OF DFL

As communication in DFL can be costly, the convergence rates of Algorithm 1 have been extensively studied. Existing studies consistently show that the topology of the decentralized graph significantly affects the convergence rates of DFL algorithms, which can be captured by the gossip matrix $\mathbf{W}$. In this section, we discuss the relationship between graph topology and the required communication rounds of DFL by examining the assumptions and the role of $\mathbf{W}$.

A. Symmetric Doubly Stochastic Matrix

The most common assumption for $\mathbf{W}$ is that $\mathbf{W}$ is symmetric and doubly stochastic [11], [21], [22].

Definition 1 (Doubly Stochastic Matrix): A matrix $\mathbf{W}$ is doubly stochastic if $\mathbf{W}$ satisfies

$$w_{ij} \geq 0, \quad w_{ij} \neq 0 \Leftrightarrow (i, j) \in \mathcal{E} \\ \mathbf{W}\mathbf{1}_m = \mathbf{1}_m, \text{ and } \mathbf{W}^\top \mathbf{1}_m = \mathbf{1}_m$$

where $\mathbf{1}_m \in \mathbb{R}^m$ is a vector of ones.

There are several properties for the symmetric doubly stochastic gossip matrix as follows.

Lemma 1: If a matrix $\mathbf{W}$ is symmetric and doubly stochastic, then

- (i). (a) All eigenvalues of $\mathbf{W}$ are real;
- (b) $|\lambda_i(\mathbf{W})| \leq 1$ with $\lambda_1(\mathbf{W}) = 1$;
- (c) $\mu(\mathbf{W}) = \|\mathbf{W} - 1/m\mathbf{1}\mathbf{1}^\top\|_{\text{op}}$, where $\|\cdot\|_{\text{op}}$ is the operator norm of a matrix.
- (ii). The state distribution of a Markov chain with the transition matrix $\mathbf{W}$ is the uniform distribution, that is, for any $\mathbf{p} \in \mathbb{R}^m$ with $\mathbf{p}^\top \mathbf{1}_m = 1$

$$\mathbf{p}^\top \mathbf{W}^t \rightarrow \frac{1}{m} \mathbf{1}_m^\top, \quad t \rightarrow \infty.$$

(iii). We have

$$\max_i \sum_j \left| (\mathbf{W}^t)_{ij} - \frac{1}{m} \right| \leq \sqrt{m} \mu^t(\mathbf{W})$$

where $(\mathbf{W}^t)_{ij}$ is the $(i, j)^{\text{th}}$ element of $\mathbf{W}^t$.

All results of Lemma 1 are well-known and studied in the context of Markov chains [23]. In particular, Lemma 1(iii) shows that the total variance difference between each row of $\mathbf{W}^t$ and the uniform distribution $1/m\mathbf{1}_m$ decreases at a rate of $\mu^t(\mathbf{W})$ as t increases. In DFL, $(\mathbf{W}^t)_{ij}$ can be interpreted as the importance of the j^{th} site to the i^{th} site after t rounds of communication. Therefore, Lemma 1(iii) plays a key role in proving the convergence of many existing DFL algorithms [11], [13], [22], and $\mu(\mathbf{W})$ can reflect the effects of graph topology on the communication rounds, as shown in Table I. Additionally, when $\mathbf{W}$ is not doubly stochastic, [24] demonstrates that additional terms will be introduced into the estimation error of the parameters. These findings highlight the significance of using the symmetric doubly stochastic matrix in DFL, which will also be a central focus of our study.

B. Fastest Mixing Markov Chain Problem

From Table I, a gossip matrix with a large μ generally requires more communication rounds than one with a small μ . Therefore, to mitigate the effects of the decentralized graph topology on convergence, the simplest approach is to minimize $\mu(\mathbf{W})$ by finding an appropriate gossip matrix $\mathbf{W}$ for $\mathcal{G}$. This approach corresponds to the fastest mixing Markov chain (FMMC) problem. Specifically, the goal of the FMMC problem is to find an optimal transition matrix $\mathbf{W}$ by solving the following constrained optimization problem:

$$\begin{aligned} \min_{\mathbf{W}} \quad & \mu(\mathbf{W}) \\ \text{s.t.} \quad & \mathbf{W}\mathbf{1} = \mathbf{1}, \mathbf{W} = \mathbf{W}^\top \\ & w_{ij} = 0 \text{ for } (i, j) \notin \mathcal{E} \\ & w_{ij} \geq \epsilon_{ij} \text{ for } (i, j) \in \mathcal{E} \end{aligned} \quad (4)$$

where ϵ_{ij} is a predefined lower bound for the weight between site i and j for each $(i, j) \in \mathcal{E}$. When each $\epsilon_{ij} = 0$, (4) becomes the standard FMMC problem extensively studied in stochastic processes. In the context of DFL, ϵ_{ij} is usually set to a small positive value to avoid assigning a zero weight to the edge between site i and j . As illustrated by [23], (4) is a convex optimization problem and is guaranteed to have at least one optimal solution, denoted by $\tilde{\mathbf{W}}$, and several algorithms have been proposed to solve it.

Definition 2 (Topology-optimal Gossip Matrix): If $\tilde{\mathbf{W}}$ is a solution of (4), then we call $\tilde{\mathbf{W}}$ a topology-optimal gossip matrix for $\mathcal{G}$.

C. Multiple Gossip Steps

Another strategy to reduce the required communication rounds of DFL is to perform MGSs within a single iteration of the algorithm. In each iteration, before the local site updates its parameters, the local estimators are communicated and aggregated $K(> 1)$ times, with K being a preset integer. The

communication and consensus step for the MGS technique in each iteration of the proposed DFL framework is outlined in the following procedure.

Communication and Consensus (MGS):

For each $i \in \mathcal{V}$, denote $\theta_i^{(t-0.5,0)} = \theta_i^{(t-0.5)}$.

- 1) For $k = 1, \dots, K$, each site sends $\theta_i^{(t-0.5,k)}$ to its neighbors and constructs

$$\theta_i^{(t-0.5,k)} \leftarrow \sum_{j \in \mathcal{N}(i)} w_{ij} \theta_j^{(t-0.5,k-1)}.$$

- 2) After K communication rounds, the new local estimator of each site i is

$$\theta_i^{(t)} \leftarrow \theta_i^{(t-0.5,K)}.$$

By straightforward algebraic manipulations, it is easy to verify that $\theta_i^{(t)}$ from MGS can be computed via

$$\theta_i^{(t)} \leftarrow \sum_{j=1}^m (\mathbf{W}^K)_{ij} \theta_j^{(t-0.5)} \quad (5)$$

which implies that using K gossip steps during each iteration of DFL is equivalent to using a “pseudo” 1GS procedure with $\mathbf{W}^K$ as the gossip matrix. Moreover, the mixing rate of $\mathbf{W}^K$ is $\mu^K(\mathbf{W})$, which decreases at an exponential rate of K since $\mu < 1$. Thus, when the MGS technique is adopted, the total number of communication rounds required is K times the convergence rate of DFL with $\mu^K(\mathbf{W})$. For instance, considering the Network-DANE algorithm included in Table I, the required communication rounds with MGS becomes

$$O\left(K \cdot \frac{\log(1/\varepsilon)}{(1-\mu^K)^2}\right) \quad (6)$$

for an arbitrary loss function. [13] show that under mild conditions, a proper selection of K can reduce the rate in (6) to as low as $O((\log(1/\varepsilon))/((1-\mu)^{1/2}))$.

Drawbacks of MGS: Although the MGS technique has demonstrated computational efficiency, it is inadequate for reducing the number of communication rounds, particularly when the decentralized graph is sparse and the gossip matrix has a large μ . In this case, $(1 - \mu^K) \approx K(1 - \mu)$ by Taylor expansion, and (6) becomes $O((c \log(1/\varepsilon))/(K(1 - \mu)^2))$. As a result, MGS improves the required communication rounds by $O(K^{-1})$. However, since $(1 - \mu)$ is very small, this improvement does not fully address the high communication cost of DFL arising from the topology of the decentralized graph.

IV. TOPOLOGY-OPTIMAL MGSS

In this section, we propose a novel MGS method aimed at mitigating the effects of the graph topology of DFL on the required communication rounds. We first introduce an ideal case for DFL to investigate the topology-optimal gossip matrix for MGS and then discuss how to achieve this gossip matrix without increasing the communication complexity in each round.

Before diving into the details, we present several notations that will be used throughout this section. We define a

K -length path between two vertices i and j in $\mathcal{G}$ as a sequence of vertices $(v_0 = i, v_1, \dots, v_{K-1}, v_K = j)$, where each pair of adjacent vertices belongs to $\mathcal{E}$, that is, $(v_{s-1}, v_s) \in \mathcal{E}$ for each $s \in \{1, \dots, K\}$. Note that some of $v_0, \dots, v_K$ may represent the same vertex. We then define $\mathcal{G}_K = (\mathcal{V}, \mathcal{E}_K)$, where $(i, j) \in \mathcal{E}_K$ if and only if there exists a K -length path between i and j . Specifically, $\mathcal{E}_K$ depicts all pairs of vertices that can share information during one iteration of the DFL framework with K gossip steps. We call $\mathcal{G}_K$ the K -round equivalent decentralized graph of $\mathcal{G}$. We use $\mathcal{W}(K)$ to denote the collection of symmetric doubly stochastic gossip matrices of $\mathcal{G}_K$. Since $(i, i) \in \mathcal{G}$ in our setting, it is straightforward to show that for any $K \leq K'$, $\mathcal{E}_K \subset \mathcal{E}_{K'}$.

A. Ideal Case on K -Round Equivalent Decentralized Graph

We begin with an ideal case to investigate the “best” consensus achievable from a topological perspective using K rounds of communication within a DFL iteration. Assume that during each communication round, each vertex can transmit all received local estimators to its neighbors. Therefore, the i^{th} vertex can acquire the current local estimator from j^{th} vertex if there exists a K -length path between them. Consequently, the communication and consensus step in the DFL framework for this ideal case is shown below.

Communication and Consensus (Ideal Case):

- 1) For $k = 1, \dots, K$, each site sends all its local estimators, including those received, to its neighbors.
- 2) Each local site updates according to

$$\theta_i^{(t)} \leftarrow \sum_{(i,j) \in \mathcal{E}_K} q_{ij} \theta_j^{(t-0.5)} \quad (7)$$

where q_{ij} is the $(i, j)^{\text{th}}$ element of a gossip matrix for a K -round equivalent decentralized graph $\mathbf{Q}$, that is, $\mathbf{Q} \in \mathcal{W}(K)$.

The topology-optimal selection of $\mathbf{Q}$ in the ideal case is described in Lemma 2 as follows.

Lemma 2 (Topology-optimal Equivalent Gossip Matrix for K Gossip Steps): If $\tilde{\mathbf{Q}}^{(K)}$ is a solution of (4) with $\mathcal{E}_K$ as the edge set and $\epsilon_{ij} = 0$, then for any $\mathbf{Q} \in \mathcal{W}(K)$, $\mu(\tilde{\mathbf{Q}}^{(K)}) \leq \mu(\mathbf{Q})$. $\tilde{\mathbf{Q}}^{(K)}$ is called the topology-optimal equivalent gossip matrix for K gossip steps.

Obviously, when a single gossip step is performed, the topology-optimal equivalent gossip matrix reduces to the topology-optimal gossip matrix, that is, $\tilde{\mathbf{Q}}^{(1)} = \tilde{\mathbf{W}}$. Lemma 2 shows that $\mu(\tilde{\mathbf{Q}}^{(K)})$ is the minimized second-largest absolute eigenvalue achieved using K communication rounds per iteration of DFL, and therefore $\mu(\tilde{\mathbf{Q}}^{(K)})$ is generally smaller than $\{\mu(\tilde{\mathbf{W}})\}^K$ as is the case for MGS.

To provide an intuitive comparison between the ideal case and MGS, and to offer insight into the interpretation of $\tilde{\mathbf{Q}}^{(K)}$, we present a toy example in Fig. 1, which illustrates the communication stages of 1GS, MGS ($K = 2$), and the ideal case. The ideal case yields a smaller second-largest absolute eigenvalue than MGS ($K = 2$) as its gossip matrix is directly derived from the two-round equivalent decentralized graph.

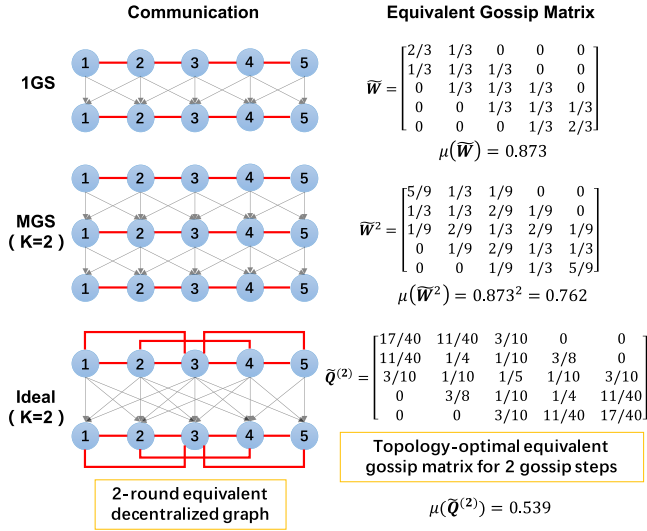


Fig. 1. Communication of 1GS, MGS ($K = 2$), and the ideal case ($K = 2$) for a chain graph with five sites, along with the equivalent gossip matrices and their second-largest absolute eigenvalues.

A more comprehensive comparison between the ideal case and MGS across a range of graph topologies is presented in Section V-A.

Though this ideal case takes full advantage of K communication rounds, the amount of information transmitted during each communication round is substantially greater than MGS. Recall that in MGS, each site only needs to transmit one local estimator to its neighbors in each communication round, while in the ideal case, each site needs to transmit all local estimators it received, resulting in significantly higher communication overhead. Therefore, we are motivated to design a new MGS method that can achieve $\tilde{Q}^{(K)}$ while maintaining the same communication complexity between each pair of connected sites as the original MGS method.

B. Gossip Tensor-Based MGSs

To develop a feasible communication-efficient MGS whose information flow approximately follows the topology-optimal equivalent gossip matrix discussed above, we propose a novel tensor-based MGS approach grounded in the following two key ideas.

- 1) Although the edges of the decentralized graph are deterministic, the gossip matrix used for each communication during MGS can vary.
- 2) During each communication, each site can transmit different contents to its neighbors.

Building on the above ideas, we introduce the concept of the gossip tensor, a novel tool to represent the communication pattern during each gossip step, offering a more flexible approach to guiding information flow compared to the static gossip matrix. A gossip tensor is an order-3 tensor of size $\mathbb{R}^{m \times m \times m}$ that generalizes the gossip matrix and allows setting the specific weights for the direct neighbors and its 2-hop neighbors of each site. We have the following definition of the gossip tensor.

Definition 3 (Gossip Tensor): The order-3 tensor $\mathbf{A} \in \mathbb{R}^{m \times m \times m}$ is called a gossip tensor of the decentralized graph $\mathcal{G}$ if $\mathbf{A}[i, j, l] = 0$ for $j \notin \mathcal{N}(i)$ or $l \notin \mathcal{N}(i)$, where $\mathbf{A}[i, j, l]$ denotes the $(i, j, l)^{\text{th}}$ element of $\mathbf{A}$.

The gossip tensor of $\mathcal{G}$ is typically sparse, with at most $\sum_{i=1}^m |\mathcal{N}(i)|^2$ nonzero entries, where $|\mathcal{N}(i)|$ denotes the cardinality of $\mathcal{N}(i)$.

We now describe how the gossip tensor is utilized in the communication and consensus step of the new method. The proposed method prepares $K - 1$ predefined gossip tensors, denoted by $\mathbf{A}^{(1)}, \dots, \mathbf{A}^{(K-1)}$, allowing for dynamically varying content during different gossip steps. Using these $K - 1$ gossip tensors, we construct a new communication and consensus step for DFL called the Tensor-based MGS (T-MGS) as follows.

Communication and Consensus (T-MGS):

For each site $i \in \mathcal{V}$, denote $\theta_{i,l}^{(t-0.5,0)} = \theta_i^{(t-0.5)}$ for each $l \in \mathcal{N}(i)$.

- 1) For $k = 1, 2, \dots, K - 1$, $\theta_{i,l}^{(t-0.5,k-1)}$ is transmitted from the i^{th} site to the l^{th} site, and the i^{th} site constructs $\theta_{i,l}^{(t-0.5,k)}$ associated with each $l \in \mathcal{N}(i)$ by

$$\theta_{i,l}^{(t-0.5,k)} \leftarrow \sum_{j \in \mathcal{N}(i)} \mathbf{A}^{(k)}[i, j, l] \theta_{j,i}^{(t-0.5,k-1)}. \quad (8)$$

- 2) At the K^{th} communication round, $\theta_{i,l}^{(t-0.5,K-1)}$ is transmitted from the i^{th} site to the l^{th} site for each $(i, l) \in \mathcal{E}$. Each site constructs the new local estimator via

$$\theta_i^{(t)} \leftarrow \sum_{j \in \mathcal{N}(i)} \theta_{j,i}^{(t-0.5,K-1)}. \quad (9)$$

The main difference between T-MGS and the traditional MGS lies in (8), where after one communication round, each site constructs a distinct local estimator for each of its neighbors, and the method for constructing the local estimator can vary across communication rounds. Specifically, $\mathbf{A}^{(k)}[i, j, l]$ describes the weight of the local estimator transmitted from the j^{th} site to the i^{th} site when constructing the local estimator to be sent from the i^{th} site to the l^{th} site in the $(k + 1)^{\text{th}}$ communication round for each $k = 1, \dots, K - 1$. In contrast, traditional MGS constructs a unified local estimator to be transmitted in the next communication round, and the method for generating the unified local estimator across different communication rounds depends on a fixed gossip matrix. An example comparing the use of the gossip matrix and the gossip tensor is illustrated in Fig. 2.

After the last communication round of T-MGS, (9) indicates that there is no need to create a weighted sum of $\theta_{j,i}^{(t-0.5,k)}$. This is because the weight of $\theta_{j,i}^{(t-0.5,k)}$ can be incorporated into $\mathbf{A}^{(K-1)}$, meaning T-MGS only requires $K - 1$ gossip tensors for K communication rounds.

The T-MGS method leaves $\mathbf{A}^{(k)} \in \mathbb{R}^{m \times m \times m}$, $k = 1, \dots, K - 1$ to be determined. Similar to (5), after simple calculations, the T-MGS method is equivalent to update $\theta_i^{(t)}$ at each i^{th} site via the weighted sum of the local estimators from sites that have

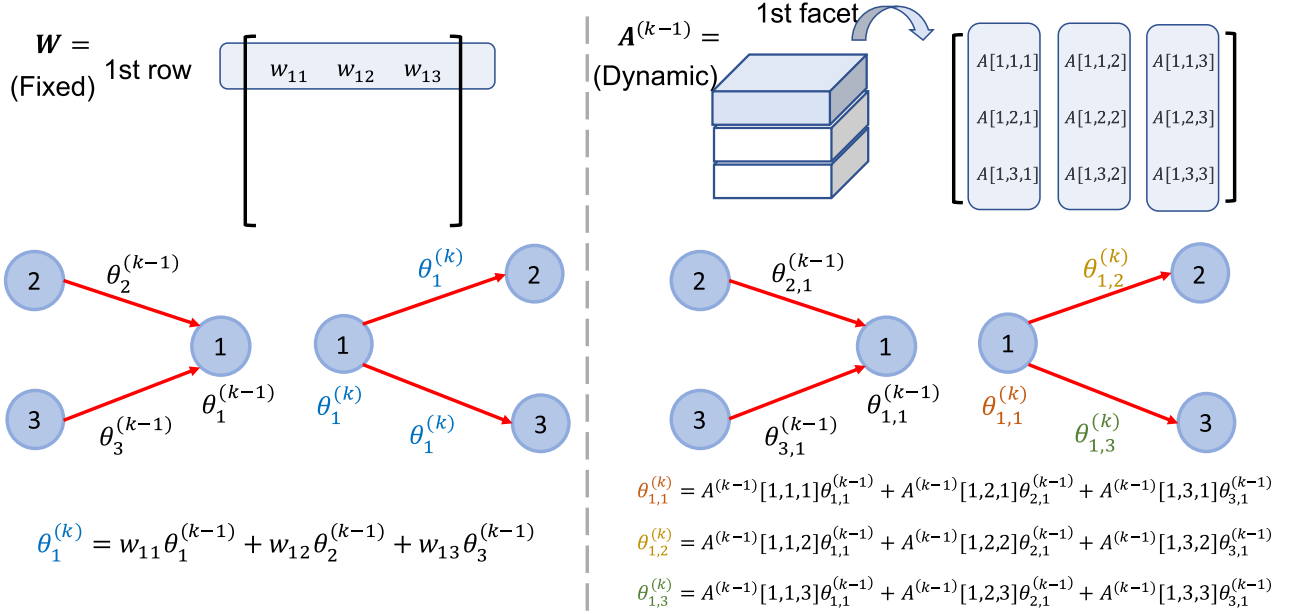


Fig. 2. Toy example of using the gossip matrix or the gossip tensor: there are three sites $\{1, 2, 3\}$ in a decentralized network with edges $(1, 2)$ and $(1, 3)$. The figure illustrates the content received by site 1 in the $(k-1)$ th communication round, as well as the content transmitted to other sites from site 1. The left panel displays the scenario using a fixed gossip matrix W , while the right panel shows the scenario using a gossip tensor $A^{(k-1)}$ specifically designed for the k th communication round.

a K -length path with the i th site

$$\theta_i^{(i)} \leftarrow \sum_{(i,j) \in \mathcal{E}_K} \dot{q}_{ij}^{(K)} \theta_j^{(i-0.5)}$$

where for each $(i, j) \in \mathcal{E}_K$

$$\dot{q}_{ij}^{(K)} = \text{CummulWeight}(i, j, K, \{\mathbf{A}^{(k)}\}_{k=1}^{K-1}).$$

The weight function $\text{CummulWeight}()$ has the following specific form:

$$\begin{aligned} & \text{CummulWeight}(i, j, K, \mathcal{G}) \\ &= \sum_{\substack{K\text{-length path } (j, v_1, \dots, v_{K-1}, i) \text{ in } \mathcal{G}}} \mathbf{A}^{(K-1)}[v_{K-1}, v_{K-2}, i] \\ & \cdot \mathbf{A}^{(K-2)}[v_{K-2}, v_{K-3}, v_{K-1}] \cdots \mathbf{A}^{(1)}[v_1, j, v_2]. \end{aligned}$$

Let $\dot{q}_{ij} = 0$ if $(i, j) \notin \mathcal{E}_K$, and define $\dot{\mathbf{Q}}^{(K)} = (\dot{q}_{ij}^{(K)}) \in \mathbb{R}^{m \times m}$. In general, $\dot{\mathbf{Q}}^{(K)}$ constructed from $\mathbf{A}^{(k)}, k = 1, \dots, K-1$ should be symmetric and doubly stochastic. This requirement poses a constraint on $\mathbf{A}^{(k)}, k = 1, \dots, K-1$. We state that there are no further constraints on the nonzero entries of $\mathbf{A}[i, j, l]$. For example, the entries of $\mathbf{A}^{(k)}$ are allowed to include negative entries.

Furthermore, if $\mathbf{A}^{(k)}, k = 1, \dots, K-1$ can make

$$\dot{\mathbf{Q}}^{(K)} = \tilde{\mathbf{Q}}^{(K)}, \quad (10)$$

then, the T-MGS method achieves the topology-optimal equivalent gossip matrix in the ideal case. In other words, the T-MGS method minimizes the effects of the graph topology and is considered topology-optimal. However, an important question arises: does there exist $\mathbf{A}^{(k)}, k = 1, \dots, K-1$ capable of achieving the topology-optimal gossip matrix? Regardless of the answer, it is essential to develop systematic methods for constructing or approximating these tensors. We will study the

approaches to determine $\mathbf{A}^{(k)}, k = 1, \dots, K-1$ in two cases: $K = 2$ and $K > 2$.

C. Determine Gossip Tensors for $K = 2$

When $K = 2$, only $\mathbf{A}^{(1)}$ needs to be determined. To achieve (10), we solve a system of $|\mathcal{E}_2|$ linear equations,

$$\sum_{l \in \mathcal{N}(i)} \mathbf{A}^{(1)}[l, j, i] = \tilde{q}_{ij}^{(2)}, \quad \text{for each } (i, j) \in \mathcal{E}_2. \quad (11)$$

Each $\mathbf{A}^{(1)}[l, j, i], l \in \mathcal{N}(i)$ appears only in the equation for $(i, j) \in \mathcal{E}_2$. Therefore, each of the $|\mathcal{E}_2|$ linear equations in (11) can be solved separately. We have Theorem 1.

Theorem 1: When $K = 2$, there always exists a gossip tensor $\tilde{\mathbf{A}}^{(1)}$ that makes (10) hold.

D. Determine Gossip Tensors for $K > 2$

Unlike the case of $K = 2$, we have Lemma 3.

Lemma 3: When $K > 2$, (10) is not guaranteed to have a solution for gossip tensors $\mathbf{A}^{(k)}, k = 1, \dots, K-1$.

Since (10) may not have an exact solution, we aim to approximate $\tilde{q}_{ij}^{(K)}$ by $\dot{q}_{ij}^{(K)}$ such that $\mu(\dot{\mathbf{Q}}^{(K)})$ is as small as possible. To this end, we determine $\mathbf{A}^{(k)}$ by solving the following constraint optimization problem:

$$\begin{aligned} & \arg \min_{\{\mathbf{A}^{(k)}\}_{k=1}^{K-1}} \mu(\dot{\mathbf{Q}}^{(K)}) \\ & \text{s.t. } \dot{q}_{ij}^{(K)} = \text{CummulWeight}(i, j, K, \{\mathbf{A}^{(k)}\}_{k=1}^{K-1}) \\ & \quad \dot{\mathbf{Q}}^{(K)} \mathbf{1} = \mathbf{1}, \dot{\mathbf{Q}}^{(K)} = (\dot{\mathbf{Q}}^{(K)})^\top \\ & \quad \dot{q}_{ij}^{(K)} \geq \epsilon_{ij} \text{ for } (i, j) \in \mathcal{E}_K \\ & \quad \mathbf{A}^{(k)}[i, j, l] = 0 \text{ if } j \notin \mathcal{N}(i) \end{aligned}$$

$$\mathbf{A}^{(k)}[i, j, l] = 0 \text{ if } l \notin \mathcal{N}(i). \quad (12)$$

However, even this optimization problem is difficult to solve in practice. Therefore, we provide a feasible approximation method that sequentially solves $K - 1$ small optimization problems. The specific steps of our method for determining gossip tensors are described in Algorithm 2. During the loop in Step 2 of Algorithm 2, we obtain one $\mathbf{A}^{(k)}$ while keeping $\mathbf{A}^{(1)}, \dots, \mathbf{A}^{(k-1)}$ fixed in each iteration. Specifically, we introduce a new series of tensors $\{\mathbf{B}^{(k)}\}_{k=1}^{K-1}$ to simplify the computation of $\dot{q}_{ij}^{(k)}$ with the CummulWeight() function, as shown in (13). In general, (13) is a convex problem with linear constraints and can be solved using convex optimization. The following theorem demonstrates that the method always provides $\hat{\mathbf{Q}}^{(K)}$ with a smaller μ than the traditional MGS.

Theorem 2: For $K \geq 3$, a solution $\hat{\mathbf{Q}}^{(K)}$ from Algorithm 2 satisfies $\mu(\hat{\mathbf{Q}}^{(K)}) \leq \mu(\hat{\mathbf{W}})^K$.

The derivatives of the constraints on $\dot{q}_{ij}^{(k)}$ and the proof of Lemma 3 and Theorem 2 are in Appendix A of the supplementary materials.

Algorithm 2 Sequential Method for Gossip Tensors

Input: K and $\mathcal{G}$.

Step 1: Construct the 2-round equivalent decentralized graph $\mathcal{G}_2$ of $\mathcal{G}$. Find $\hat{\mathbf{Q}}^{(2)}$ by solving (4). Then, solve (11) with $\hat{\mathbf{Q}}^{(2)}$ to obtain one solution as $\hat{\mathbf{A}}^{(1)}$. Let $\mathbf{B}^{(1)} = \hat{\mathbf{A}}^{(1)}$.

Step 2:

for $k = 3, \dots, K$ **do**

1. Set $\mathbf{B}^{(k-1)} \in \mathbb{R}^{m \times m \times m} = \mathbf{0}$. For each $(i, j) \in \mathcal{E}_k$, $v_{k-1} \in \mathcal{N}(i)$, $v_{k-2} \in \mathcal{N}(v_{k-1})$, set

$$\mathbf{B}^{(k-1)}[v_{k-2}, j, v_{k-1}] \leftarrow \sum_{v_{k-3} \in \mathcal{N}(v_{k-2})} \hat{\mathbf{A}}^{(k-2)}[v_{k-2}, v_{k-3}, v_{k-1}] \mathbf{B}^{(k-2)}[v_{k-3}, j, v_{k-2}].$$

2. Obtain $\hat{\mathbf{A}}^{(k-1)}$ from a solution of

$$\begin{aligned} & \arg \min_{\mathbf{A}^{(k-1)}} \mu(\hat{\mathbf{Q}}^{(k)}) \\ & \text{s.t.} \quad \dot{q}_{ij}^{(k)} = \sum_{v_{k-1} \in \mathcal{N}(i)} \sum_{v_{k-2} \in \mathcal{N}(v_{k-1})} \mathbf{A}^{(k-1)}[v_{k-1}, v_{k-2}, i] \mathbf{B}^{(k-1)}[v_{k-2}, j, v_{k-1}], \\ & \quad \hat{\mathbf{Q}}^{(k)} \mathbf{1} = \mathbf{1}, \hat{\mathbf{Q}}^{(k)} = (\hat{\mathbf{Q}}^{(k)})^\top, \\ & \quad \dot{q}_{ij}^{(k)} \geq \epsilon_{ij}^{(k)} \text{ for } (i, j) \in \mathcal{E}_k \\ & \quad \mathbf{A}^{(k-1)}[i, j, l] = 0 \text{ if } j \notin \mathcal{N}(i), \\ & \quad \mathbf{A}^{(k-1)}[i, j, l] = 0 \text{ if } l \notin \mathcal{N}(i). \end{aligned} \quad (13)$$

end for

Step 3. Calculate $\hat{\mathbf{Q}}^{(K)}$ by

$$\hat{q}_{ij}^{(K)} = \text{CummulWeight}\left(i, j, K, \{\hat{\mathbf{A}}^{(k)}\}_{k=1}^{K-1}\right)$$

Output: $\hat{\mathbf{A}}^{(k)}, k = 1, \dots, K - 1$, and $\hat{\mathbf{Q}}^{(K)}$.

Remark: We note that the solutions to (11) are generally not unique. Since Algorithm 2 follows a sequential approach, variations in the solutions at earlier steps will influence the outcomes of later steps, resulting in different values for $\hat{\mathbf{Q}}^{(K)}$.

In practice, we can run Algorithm 2 multiple times and select the set of $\hat{\mathbf{A}}^{(k)}, k = 1, \dots, K - 1$ that yields the smallest $\hat{\mathbf{Q}}^{(K)}$.

E. T-MGS for DFL

In this subsection, we summarize the procedure for applying T-MGS within DFL and analyze its complexity. The practical implementation of T-MGS involves the following steps.

- 1) *Gossip Tensor Construction:* Given a deterministic decentralized graph $\mathcal{G}$ and a predefined number of gossip steps K , construct $K - 1$ gossip tensors $\hat{\mathbf{A}}^{(k)}$ for $k = 1, \dots, K - 1$ using Algorithm 2.
- 2) *System Setup:* For the i^{th} site in the DFL system, store the sparse matrix slice $\hat{\mathbf{A}}^{(k)}[i, :, :]$ of each gossip tensor $\hat{\mathbf{A}}^{(k)}, k = 1, \dots, K - 1$ locally, which includes the communication weight relevant to that site for each gossip step.
- 3) *DFL Execution:* Execute Algorithm 1 with 1GS replaced by the proposed T-MGS communication and consensus to obtain the final estimate of θ at each site.

A step-by-step example of applying T-MGS with $K = 2$ in DFL can be found in Appendix B of the supplementary materials.

We now discuss the complexity of T-MGS from three aspects, comparing it to the original MGS method.

- 1) *Communication Complexity:* The communication complexity of T-MGS is identical to that of the original MGS. In both methods, during each communication round, every pair of connected sites exchanges an estimate of θ . This is a major benefit of T-MGS that it achieves enhanced learning performance without any additional communication cost.
- 2) *Storage Complexity:* In the original MGS method, each site only needs to store d_i weights, where d_i is the degree of the i^{th} site. In contrast, T-MGS requires each site to store sparse matrix slices $\hat{\mathbf{A}}^{(k)}[i, :, :]$ for $k = 1, \dots, K - 1$. Each slice contains approximately d_i^2 nonzero entries. As a result, T-MGS introduces an additional storage cost of $O(K \max(d_i^2))$ per site. This increase remains manageable, particularly in sparse graphs.
- 3) *Computational Complexity:* T-MGS with K gossip steps requires solving (4) and (11) one time and solving (13) $K - 2$ times. As mentioned earlier, (11) can be divided into several small linear systems that can be solved efficiently. (4) and (13) are both semidefinite programming (SDP) problems, a type of convex optimization that can be efficiently solved using standard methods. For example, we employed the splitting conic solver (SCS) [25] and provided an efficient implementation of the proposed gossip tensor construction method, available on the authors' GitHub. The computational complexity of SCS is around $O(m^3/\delta)$ to achieve δ accuracy. Runtime evaluations for our implementation are presented in Appendix C of the supplementary materials. During DFL execution, T-MGS requires the i^{th} site to construct d_i linear combinations of the received estimators of θ in each communication round, while the original MGS constructs only one. This additional computational cost

is minor and can be efficiently handled by modern computing systems.

V. NUMERICAL EXPERIMENT

In this section, we evaluate the performance of the proposed method in reducing the communication rounds of DFL from two aspects. First, we compare the second-largest absolute eigenvalues of different communication and consensus steps. Second, we investigate the performance of various methods in DFL problems using simulated and real datasets.

We consider six representative graph typologies for the decentralized graph, as shown in Fig. 3, which includes examples of these six typologies with $m = 10$.

- 1) *Ring*: The ring graph is a fundamental structure used in decentralized computing, where the i^{th} vertex in the graph is connected with the $(i + 1 \bmod m)^{\text{th}}$ vertex.
- 2) *Binary Tree*: A binary tree graph is the simplest hierarchical graph structure where each node has at most two children and two nodes are connected if and only if one site is the parent of the other. There is only one path between any two nodes in the binary tree.
- 3) *Grid*: A 2-D $n_1 \times n_2$ grid graph is a graph Cartesian product of path graphs on n_1 and n_2 vertices. We fix $n_1 = 5$ and set $n_2 = m/n_1$ to maintain the local sparsity during the expansion of the network.
- 4) *Torus*: The torus grid graph is formed from the graph Cartesian product of two ring graphs with n_1 and n_2 nodes, respectively. It can be naturally embedded on the surface of a torus with no edge intersections. We also fix $n_1 = 5$ and set $n_2 = m/n_1$ in the experiment.
- 5) *Fixed Degree*: Each node is linked to a fixed number D of sites in the fixed degree graph. We use $D = 3$.
- 6) *Erdős-Rényi (ER)*: ER graph is a random graph where each pair of nodes has a prespecified probability of p to connect. We generated a connected ER graph with $p = 3/m$.

A. Comparing the Second-Largest Absolute Eigenvalue of Different Methods

Recall that $\tilde{\mathbf{W}}^K$, $\tilde{\mathbf{Q}}^{(K)}$, and $\dot{\mathbf{Q}}^{(K)}$ are the equivalent gossip matrices for traditional MGS, the ideal case, and T-MGS with K gossip steps. We compute $\mu(\tilde{\mathbf{W}}^K) = \mu(\tilde{\mathbf{W}})^K$, $\mu(\tilde{\mathbf{Q}}^{(K)})$, and $\mu(\dot{\mathbf{Q}}^{(K)})$ for $K = 2, 4$ across varying values of m from 10 to 50 for six distinct topologies, with $\mu(\tilde{\mathbf{W}})$, represented by *IGS* as a baseline. For clarity and brevity, we denote them as *MGS-K* = 2, 4, *Ideal Case-K* = 2, 4, and *T-MGS-K* = 2, 4, respectively. Since the fixed degree and ER graphs are not unique for each m , we generate multiple instances and report the mean values.

The comparison results are presented in Fig. 4. Across all graph topologies and values of m , both the ideal case and T-MGS achieve a lower μ compared to conventional MGS with the same value of K , with the gap being larger for $K = 4$ than for $K = 2$. This result shows that the proposed T-MGS can significantly decrease μ without increasing the communication complexity of traditional MGS, thereby accelerating the convergence rate. Comparing μ of the ideal case and T-MGS, we observe that the curves for the ideal case and T-MGS with

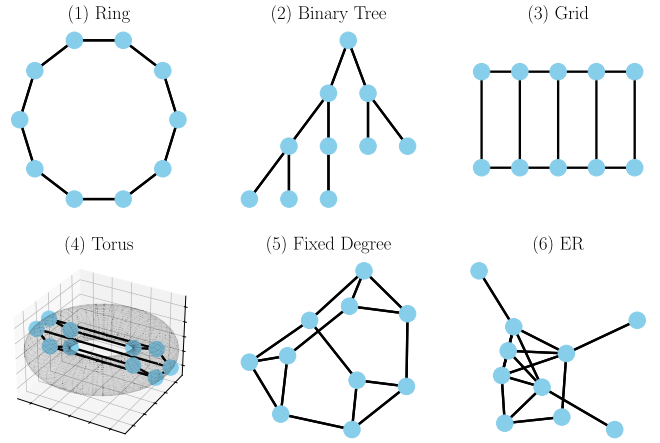


Fig. 3. Illustration of six graph topologies with $m = 10$.

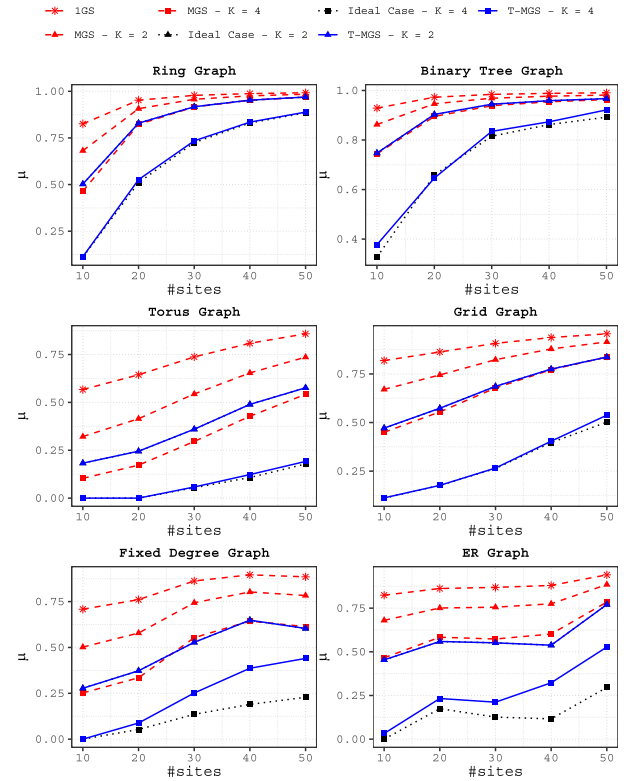


Fig. 4. Comparison of μ for MGS, the ideal case, and T-MGS.

$K = 2$ overlap exactly, reflecting Theorem 1. When $K = 4$, $\mu(\dot{\mathbf{Q}}^{(K)})$ remains close to $\mu(\tilde{\mathbf{Q}}^{(K)})$ in most cases, suggesting that our proposed algorithm effectively approximates the ideal case in terms of the mixing rate. Although, for complex topologies such as fixed-degree and ER graphs, the proposed T-MGS algorithm has a relatively larger μ than the ideal case for large m , we will demonstrate in the following numerical experiments that it still significantly outperforms MGS.

B. Experiments on Simulated Data and Real Data

We now evaluate the performance of different methods on simulated and real datasets for three different models—linear regression, logistic regression, and CNN.

1) *Simulation on Linear Regression and Logistic Regression*: For both linear regression and logistic regression, we

constructed simulated datasets. We set $m = 20$ and generated 1000 observations at each site, with ten features sampled from a standard normal distribution. The regression coefficients for these ten features were randomly sampled from a uniform distribution $U[0, 1]$. The response variables for linear regression and logistic regression were generated sequentially. Additionally, 1000 observations were generated as the test dataset to evaluate model performance.

To demonstrate the performance of T-MGS, we included five comparative methods: the 1GS method, the classical MGS with $K = 2, 4$, and the proposed T-MGS with $K = 2, 4$. To compare the methods, we consider: 1) estimation performance by the mean parameter estimated error over m sites and 2) the prediction performance by the mean relative prediction error for linear regression and the classification rate for the logistic regression in the test dataset. We selected the Network-DANE algorithm [13] due to its good convergence properties for both linear and logistic regression models.

The performance of different methods across six decentralized graph topologies for linear and logistic regressions is presented in Figs. 5 and 6, respectively. We analyze how performance measures change with the number of iterations and communication rounds. The results for the two models are very similar. First, regarding the number of iterations, T-MGS demonstrates promising improvements. When $K = 4$, it achieves the fastest convergence rate across all considered topologies. Even with $K = 2$, it exhibits rapid convergence that is comparable to MGS with $K = 4$ in relatively sparse topologies, such as ring, binary tree, and fixed degree graphs. In terms of communication rounds, T-MGS with $K = 4$ maintains superior performance in sparse topologies. For example, to achieve a specific relative loss difference in the ring graph, the traditional MGS requires significantly more communication rounds compared to T-MGS with $K = 4$. Specifically, in the ring graph and the logistic regression, while the traditional MGS needs more than 200 rounds of communication to reach a classification rate of 0.94, T-MGS with $K = 4$ requires only 60 rounds. Although T-MGS with $K = 2$ performs best on grid and torus graphs due to their higher connectivity, this advantage diminishes and T-MGS with $K = 4$ performs best when $m = 100$ and all graphs become sparser, as shown in Appendix D of the supplementary materials. These results demonstrate the efficiency of the proposed T-MGS method in mitigating the effects of graph topologies in DFL and reducing the overall number of communication rounds required.

2) *CIFAR-10 Real Dataset by CNN*: We used the CIFAR-10 dataset for the CNN model, which consists of 60 000 color images of size 32×32 pixels, categorized into ten classes. The dataset is divided into 50 000 training images and 10 000 testing images, with the training set split equally into $m = 60$ vertices. We consider both homogeneous and heterogeneous sites. In the homogeneous case, the training set is split randomly, while in the heterogeneous case, each site has more observations in two classes, with the class ratio being $2.5 : 1.5 : 1 : \dots : 1$. The CNN architecture consists of two convolutional layers, each followed by batch normalization, ReLU activation, and dropout (with a probability of 0.1). We calculated the classification rate on the test dataset to evaluate

the performance of different methods during the iterations. We used the DFedAvgM algorithm [13] due to its favorable convergence performance for CNN.

The performance of different methods across six decentralized graph topologies is shown in Fig. 7. The results align with those from the simulations. In all topologies and both cases, the proposed T-MGS with $K = 4$ or $K = 2$ required the fewest communication rounds among all compared methods to achieve high prediction accuracy. These findings further demonstrate the efficiency of the proposed method. Since the simulation and real data experiments are based on different DFL algorithms, the consistent results indicate that the proposed method is applicable across various DFL algorithms, effectively reducing the required communication rounds. More experimental results can be found in Appendix D of the supplementary materials.

VI. RELATED WORK

There are several groups of work related to our article.

A. Decentralized Computing

Decentralized computing algorithms address the model learning problem with data scattered across multiple sites, with their interrelations represented by a graph. As early as 2009, [20] proposed decentralized gradient descent (DGD), one of the simplest and most efficient decentralized algorithms. A series of studies have worked on the convergence rate of DGD in both convex and nonconvex optimization [26], [27]. Later on, many first-order decentralized algorithms are proposed such as EXACT [11], [28], [29], and [30] studied decentralized stochastic gradient descent (DSGD). [31], [32], and [33] studied the gradient tracking technique (also called momentum) in decentralized computing. [12] and [34] designed proximal gradient algorithms for decentralized computing. [13] proposed several approximate Newton-type methods for decentralized computing. [35] studied the optimal computation and communication complexity for the decentralized convex optimization problem. Most of these decentralized computing algorithms can be applied to solving the DFL problem after considering the real application constraints in DFL.

B. Decentralized FL

DFL imposes higher demands on data privacy and communication efficiency. Specifically, the original data in each site is not allowed to be transmitted to the other site, and the communication between sites can be costly. [6] took a Bayesian-like DFL approach via the introduction of a belief over the model parameter space. [7] proposed the decentralized federated averaging with momentum (DFedAvgM), which includes multiple local training before communication to save communication costs. As DFL can also face the problem of heterogeneous data among sites, several heterogeneous-aware works are made on DFL [36], [37]. A systemic review of existing DFL studies can be found in [38].

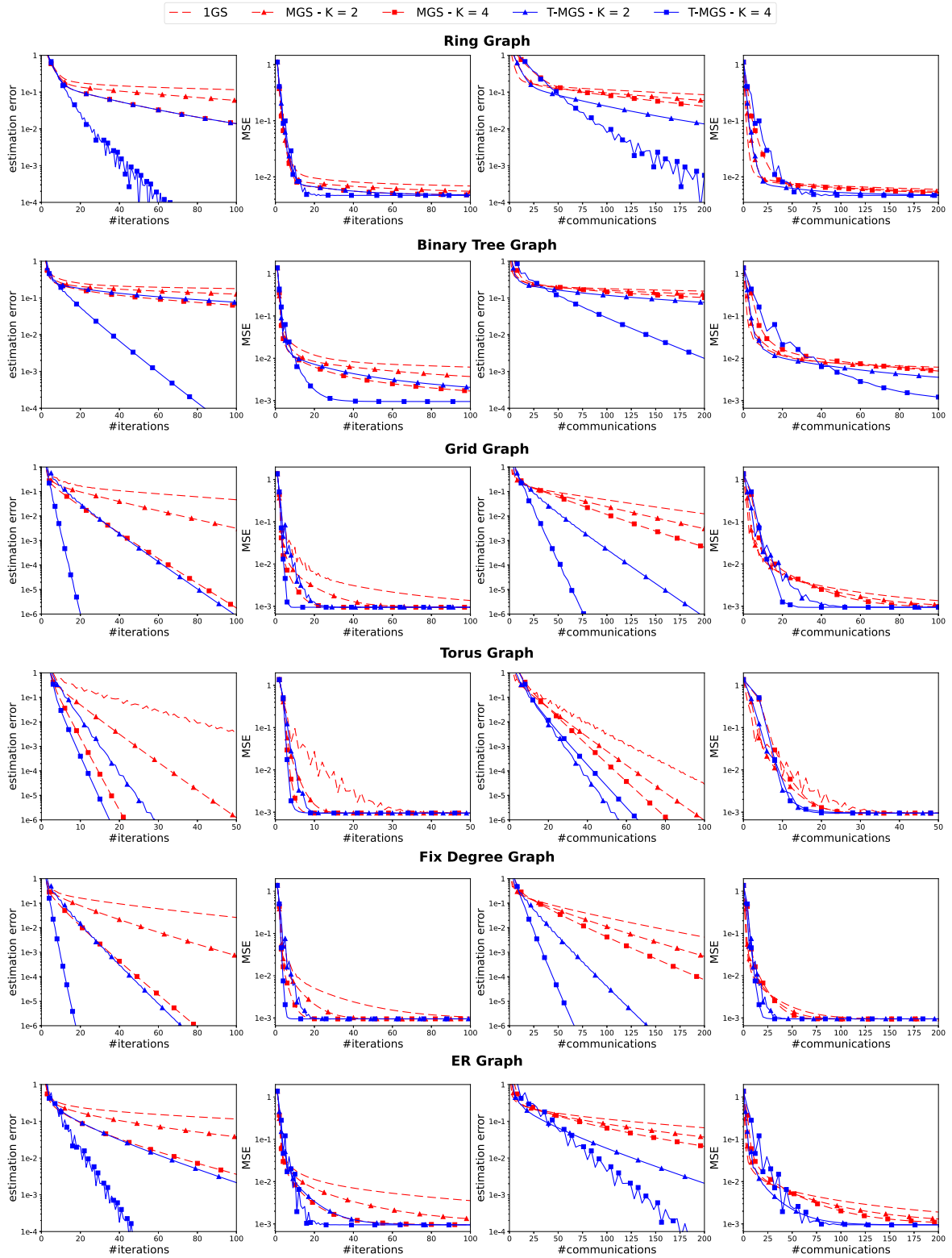


Fig. 5. Performance of different methods in the linear regression model with simulated datasets.

C. Multiple Gossip Steps

MGS refers to the technique of processing multiple rounds of communication before updating parameters during the DFL. In general, increasing the number of gossip steps allows sites to receive information from far-apart sites during each iteration and thereby achieve a more consensus model similar

to that seen in centralized FL. Several works have shown that using MGS can theoretically improve the convergence properties of DFL. For example, [32] proposed FastMix with MGS and gradient tracking and shows that using MGS can further improve the convergence rate. [18] emphasized the theoretical benefits of MGS in DGD. [13] proposed

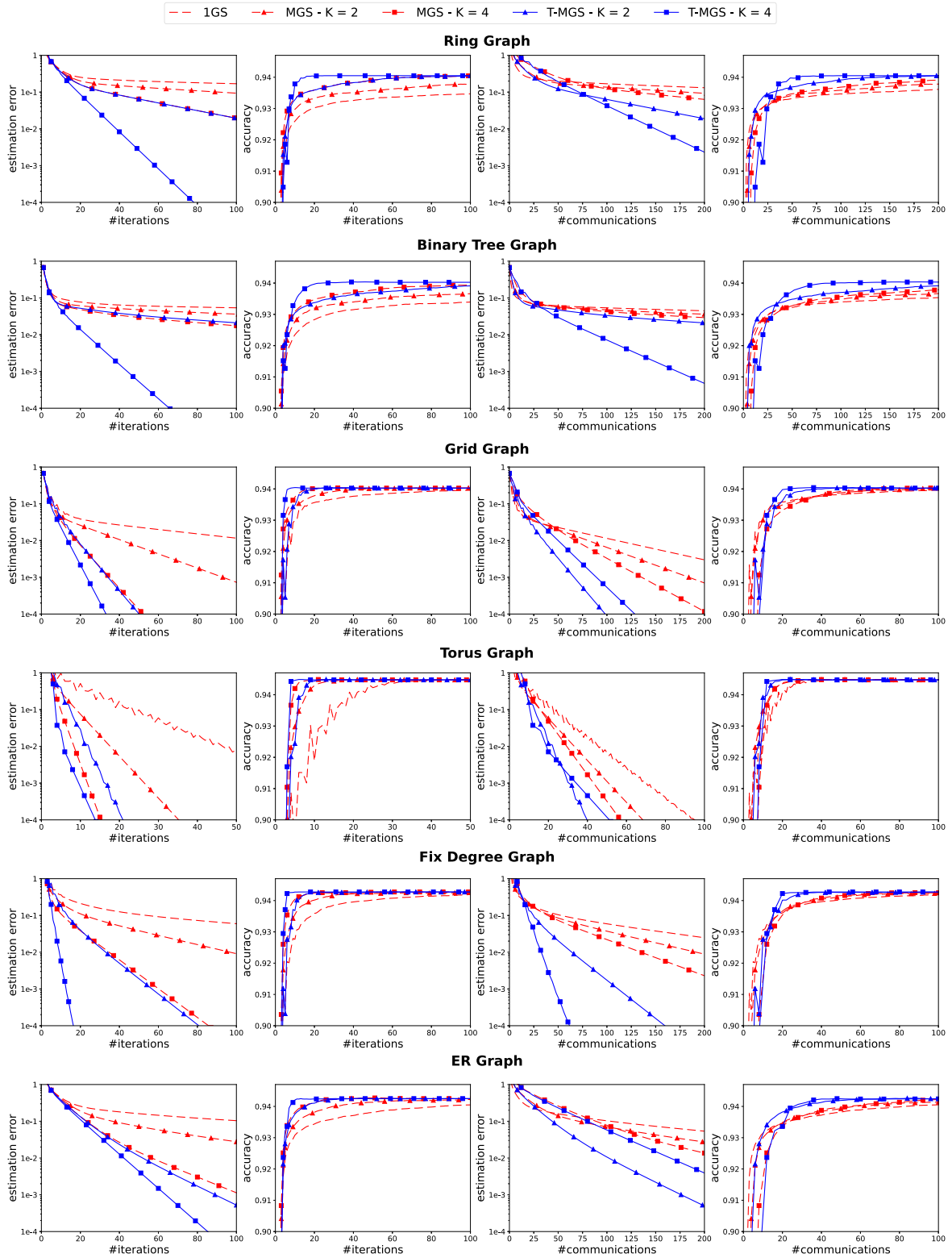


Fig. 6. Performance of different methods in the logistic regression model with simulated datasets.

Network-DANE with MGS, and [39] proposed a decentralized exact efficient PCA algorithm with MGS. Both works show that MGS can help achieve a faster convergent rate. [40] designed a flexible DFL strategy with MGS and multiple local updates, which can optimize the convergence rate of DFL by balancing the communication and computing costs.

[19] adopted MGS to accelerate the aggregation of local flat models and achieve better model consistency of DFL. More recently, [41] established the first convergence lower bound for decentralized learning over row-stochastic networks by incorporating an MGS strategy to address instability issues. [42] provided a unified stability-based analysis showing that

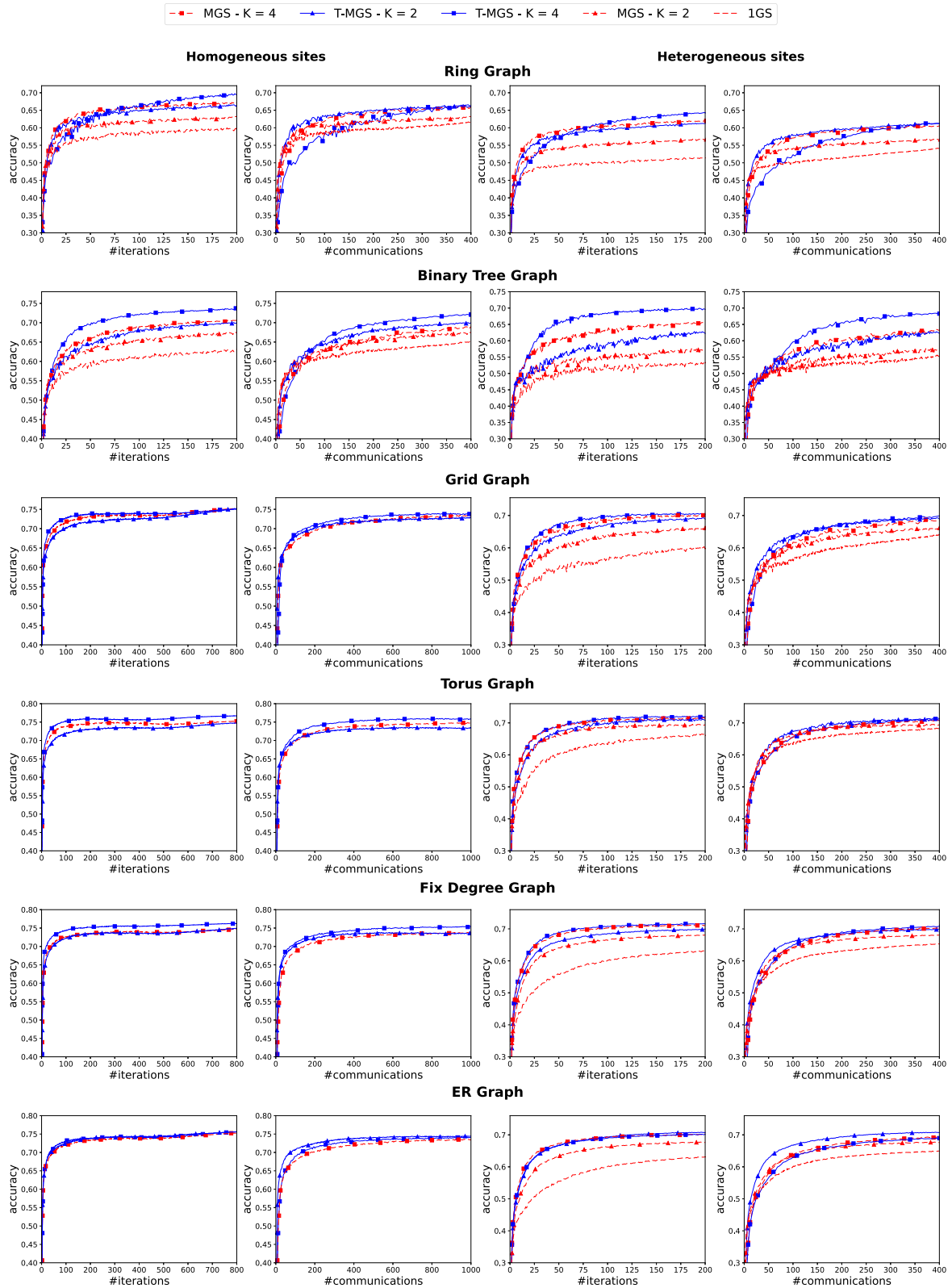


Fig. 7. Performance of different methods in CNN with CFAR10. The left two columns are for the homogeneous sites, and the right two columns are for the heterogeneous sites.

MGS can reduce the optimization error and generalization bound at an exponential rate but still leaves a nonnegligible gap from the centralized setting. [43] proposed an MG-

skipping method that probabilistically skips multigossip communications in most iterations under strongly convex settings to further reduce communication costs.

Compared with the proposed T-MGS, all these existing MGS studies employ a fixed mixing weight matrix during MGSs.

D. Dynamic Gossip Learning

Dynamic gossip learning refers to a class of DFL methods that utilize different communication graphs across iterations. The core idea is to use a sequence of much sparser subgraphs derived from the original static graph to reduce communication costs in each round while maintaining the same convergence rate. For instance, [14] introduces one-peer exponential graphs and shows that when the number of sites is a power of two, these sparse graphs can match the performance of the original dense topology while significantly reducing communication overhead. Later, [16] and [44] proposed DSGD-CECA and EquiTopo, respectively, to achieve similar properties for arbitrary numbers of sites. [17] proposed Base- $(k + 1)$ graph to achieve both a fast consensus rate and a small maximum degree. Although these dynamic methods share similarities with T-MGS, there are important differences. Both approaches vary their operations across communication rounds. However, dynamic gossip learning focuses on reducing communication by selecting a subset of connected sites to participate in each round, effectively lowering the total communication cost. In contrast, T-MGS retains full communication across all connected sites but introduces a site-specific design for the transmitted content, allowing T-MGS to preserve a rich information flow and improve learning performance.

VII. CONCLUSION AND DISCUSSION

In this article, we introduce a novel MGS method designed to reduce the number of communication rounds in DFL from a topological perspective. By utilizing gossip tensors to guide the transmission of local information, the proposed method minimizes the effects of decentralized graph structures and approximates the topology-optimal MGS scenario, in which all received information is effectively transmitted to neighbors during each communication stage. Furthermore, we present a series of efficient optimization problems to compute these gossip tensors. Experiments on both simulated and real-world datasets demonstrate the effectiveness of T-MGS.

There are several limitations and future directions for T-MGS. First, T-MGS is suited for sparse graphs (including the extremely sparse graph) with a low second-largest absolute eigenvalue and significantly outperforms the original MGS. However, for dense matrices, the improvement of T-MGS is relatively small because the decentralized topology of dense graphs (i.e., high node degree) closely resembles a centralized one, allowing information to flow directly between most nodes with minimal loss, thus leaving less room for improvement over MGS. For practical guidance on T-MGS, researchers can preestimate its benefits for a graph using μ before DFL. Similar to the comparison in Fig. 4, we can compute $\mu(\mathbf{Q}^{(K)})$ and $\{\mu(\tilde{\mathbf{W}})\}^K$ and compare if there is a significant decline from $\{\mu(\tilde{\mathbf{W}})\}^K$ to $\mu(\mathbf{Q}^{(K)})$ and investigate its impact on the convergence rate of the selected DFL algorithm. This μ -based method allows for assessing the effects of T-MGS on any graph beyond the six topologies evaluated in the article (e.g., the

planar graph [45] and graphs generated by the preferential attachment model [46], ensuring its practical applicability.

Second, as a starting point, the current T-MGS focuses primarily on graph topology and does not explicitly integrate graph design with local properties such as data heterogeneity or personalization. Recently, a concurrent work [47] also explored the idea of varying gossip weights and proposed an adaptive one-step gossip strategy based on local model discrepancies. However, their approach does not generalize to MGS settings and does not explicitly incorporate network topology. Since T-MGS is a flexible framework that allows varying weights and transmitting different contents to neighbors, a promising future direction is to integrate such adaptive information into the gossip tensor, making it applicable to both graph-structural and local-model information.

Third, another promising direction for future research is the integration of T-MGS and dynamic gossip learning. Specifically, one could jointly optimize the selection of transmitting sites and the design of their transmitted content. By balancing communication efficiency with convergence rates, such a hybrid strategy could reduce communication costs while mitigating the influence of network topology.

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